

Bellabeat

Bellabeat is a wellness tech company that wants to understand how people use smart fitness devices. The goal of this project is to analyze Fitbit user activity data to detect patterns in daily activity and generate insights that can help inform Bellabeat's marketing strategy.

In this analysis, we use the dataset Fitbit Fitness Tracker Data from Kaggle. The dataset consists of Fitbit user activity and wellness data. This data includes daily steps, distance traveled, calories burned and activity intensity levels. The data is available in the public domain under CC0 and is used for educational purposes.

Business Challenge

The business problem is to analyse Fitbit user activity data to detect trends of physical activity of users and to understand how they are using their health and fitness tracking devices. The objective is to apply these findings to find opportunities for Bellabeat to enhance user engagement and encourage healthier daily activity habits.

Questions for Analysis

- Is there a link between daily steps and calories burned?
- What is the users' activity pattern on different days of the week?
- How long do users spend at various activity levels?
- How are the recorded days distributed in terms of daily step counts?
- What can Bellabeat do to leverage this information for increased physical activity?

Data Source & Preparation

This analysis uses Fitbit fitness-tracking data containing information about users' daily activity, including steps, calories burned, activity minutes, distance, and activity dates.

The data was prepared and analysed using **Tableau Public**. The dataset was imported into Tableau and the relevant activity measures were used to create visualisations examining relationships, daily activity patterns, activity intensity, and the distribution of daily step counts.

The analysis focuses on the following measures:

- Total Steps
- Calories
- Very Active Minutes
- Fairly Active Minutes
- Lightly Active Minutes
- Sedentary Minutes

- Activity Date

Objective

The overall objective is to utilize fitness-tracking data to understand customer's behaviour and provide data-driven suggestions that can help Bellabeat better engage its users and promote healthier activity.

Data Analysis & Findings

The Fitbit activity data was analysed using Tableau Public to identify patterns in daily activity, step counts, calorie usage, and activity intensity. Four visual graphs were created to investigate the key analysis questions.

1. Steps vs Calories

The analysis shows a generally positive relationship between daily steps and calories burned. In the dataset, days with higher step counts generally corresponded with higher calorie expenditure.

Key finding: Increasing daily movement appears to be associated with greater calorie expenditure among the recorded Fitbit users.

2. Average Steps by Weekday

Average daily steps were compared across the seven days of the week. **Saturday had the highest average step count, while Sunday had the lowest.**

Key finding: Activity levels varied throughout the week, with the highest average activity occurring on Saturday and the lowest on Sunday.

3. Average Activity Minutes

The analysis compared the average amount of time users spent at different activity levels. Users recorded an average of **991.2 minutes of sedentary time per day**, compared with **192.8 minutes of light activity, 21.2 minutes of very active time, and 13.6 minutes of fairly active time.**

Key finding: Sedentary time was substantially higher than the other activity categories, suggesting that encouraging more everyday movement could be an important opportunity for Bellabeat.

4. Daily Step Distribution

The distribution of daily step counts showed that relatively low-step days were common within the dataset. There were **109 recorded days with fewer than 1,000 steps**, while the next three step ranges contained 82, 70, and 71 recorded days respectively.

Key finding: Low-step days were common among the recorded activity data. However, very low step counts should be interpreted cautiously because they may represent days when a Fitbit device was not worn or did not fully record activity.

Overall Finding

All four analyses suggest that consistent daily movement may be more important than focusing only on intense exercise. Users logged sedentary time at a significantly higher rate than active time. Low-step days were also common in the dataset.

Instead of only focusing on intense workouts, it allows Bellabeat to focus on marketing strategies and features that promote small, incremental increases in physical activity every day.

Recommendations

Based on the analysis of Fitbit user activity patterns, the following recommendations could help Bellabeat improve user engagement and encourage more consistent physical activity.

1. Encourage movement throughout the day

The data revealed that users recorded significantly more sedentary time than active time. Bellabeat has the opportunity to create personalised reminders to encourage users to stand, walk or do short physical activities after long periods of inactivity.

2. Promote consistent daily activity

Low-step days were common in the dataset. Bellabeat could introduce personalised daily step goals and progress notifications to encourage users to gradually increase their activity rather than focusing only on intense exercise.

3. Use weekly activity patterns for engagement

Saturday had the highest average daily step count, while Sunday had the lowest. Bellabeat could use this pattern to create targeted weekend engagement, such as Sunday activity reminders or short weekend movement challenges.

4. Use positive reinforcement

The positive relationship between steps and calories burned suggests that increasing daily movement is associated with greater calorie expenditure. Bellabeat could use progress tracking, milestones, badges, or personalised messages to show users the benefits of maintaining their daily activity.

Overall Recommendation

Bellabeat should focus on the importance of small, gradual adjustments in daily movement rather than the need of high-intensity exercise alone. Personalised reminders, achievable step goals, progress tracking and short activity challenges could support users to reduce sedentary behaviour and develop more consistent activity habits.

Limitations

There are several limitations to consider when interpreting the results of this analysis.

1. Limited sample size

The dataset contains activity records from a relatively small group of Fitbit users. Therefore, the findings may not represent the behaviour of the wider population of fitness-tracking users or Bellabeat's customers.

2. Potentially incomplete tracking

Very low step counts may indicate that a user was genuinely inactive, but they could also indicate that the Fitbit device was not worn or did not record activity for the entire day. This means very low activity values should be interpreted cautiously.

3. Observational data

The analysis identifies relationships and patterns in the data but does not establish cause and effect. For example, the positive relationship between steps and calories burned does not prove that increasing steps alone will cause a specific increase in calorie expenditure.

4. Limited user context

The dataset provides activity measurements but does not provide enough information about users' individual circumstances, such as their lifestyle, occupation, fitness goals, or reasons for being active or inactive. These factors could influence their activity patterns.

Overall limitation

The findings should therefore be viewed as **directional insights rather than definitive conclusions about all fitness-tracking users**. Further analysis using a larger and more representative dataset, as well as additional user demographic and behavioural information, could provide stronger evidence for Bellabeat's marketing decisions.

Conclusion

This analysis of Fitbit user activity data identified several patterns that may provide opportunities for Bellabeat to improve user engagement. Daily steps showed a positive relationship with calories burned, while activity levels varied across the week, with Saturday recording the highest average daily steps and Sunday the lowest.

The most significant pattern was the large amount of sedentary time recorded compared with time spent in higher-intensity activities. The distribution of daily steps also showed that low-step days were common within the dataset.

Based on these findings, Bellabeat could focus on encouraging **consistent everyday movement** through personalised reminders, achievable activity goals, progress tracking, and short movement challenges. While the dataset has limitations and the findings should not be generalised to all users, the analysis provides useful directional insights that could inform Bellabeat's user engagement and marketing strategies.